System Architecture & Technical Design Document

AI-Powered Restaurant Operations & Review Assistant — Engineering Specification

1. Problem Statement (Business Problem Addressed)

Multi-unit restaurant enterprises manage thousands of unstructured customer reviews across Yelp and delivery aggregators. Traditional sentiment analysis delivers high-level vanity metrics (e.g., '76% positive') without diagnosing the operational root causes of service friction (e.g., Friday 8:30 PM expo heat-lamp staging delays or patio drink refill backlogs). High-severity incidents remain undetected until customer churn occurs. General managers lack automated tools linking customer complaints to prioritized Standard Operating Procedures (SOPs) with defined Service Level Agreement (SLA ≤ 2h) response windows.

2. Architecture Design & End-to-End Pipeline Flow

The system executes a cohesive 10-stage end-to-end pipeline: **Dataset** → **Preprocessing** → **Chunking** → **Embeddings** → **Vector Database** → **Retriever** → **LLM / RAG Pipeline** → **Agents** → **API Layer** → **User Interface**. Offline Ingestion prepares 48,288 reviews via chunking and MiniLM embeddings persisted into ChromaDB and SQLite. In the online serving flow, user inquiries pass through FastAPI security guardrails, trigger dense ChromaDB and sparse BM25 hybrid retrieval fused via Reciprocal Rank Fusion (k=60), augment Gemini 2.5 LLM prompts, execute LangGraph's 5-agent state graph, and render interactive analytics on Streamlit.

3. Technology Selection & Technical Justifications

Component	Selected Tech	Alternatives Considered	Engineering Justification
Primary LLM	Google Gemini 2.5 Flash / Pro	GPT-4o, Claude 3.5	1M+ context window, JSON Schema compliance, 100+ t/s throughput, $0.075/1M tokens, grounded fallback.
Embedding Model	all-MiniLM-L6-v2 (384-d)	text-embedding-3-small	Zero API cost/network latency (<10ms CPU inference), 75% RAM savings over 1536-d, complete data privacy.
Vector Database	ChromaDB (v0.4.22+)	Pinecone, Milvus, Weaviate	Embedded in-process storage, HNSW cosine index, zero external cluster management, native metadata filters.
API & Web Frameworks	FastAPI + Streamlit	Django / Flask + React	Asynchronous ASGI throughput, auto OpenAPI docs, rapid interactive analytics with native Plotly.
Agent & Forecasting	LangGraph + Facebook Prophet	CrewAI + ARIMA/LSTM	Cyclic state graph with critic judge; Prophet models weekly dining rushes/holidays with 95% CI on CPU.

4. Retrieval Strategy (Chunking, Embedding & Hybrid RRF)

• **Chunking Strategy:** Whole-review document level chunking (mean length ~135 tokens). Preserves complete dining narratives without cross-sentence fragmentation.
• **Embedding Approach:** L2-normalized 384-dimensional dense vectors generated by all-MiniLM-L6-v2 in Cosine Similarity space.
• **Hybrid Search (RRF):** Executes parallel dense ChromaDB (top-50) and sparse BM25 (top-50), merged via Reciprocal Rank Fusion: $RRF_Score(d) = \Sigma\ 1 / (60 + rank_m(d))$. Eliminates keyword vs semantic blind spots with a direct SQLite fallback for small sample sizes.

5. Agent Design (Roles, Workflow & State Machine Communication)

LangGraph StateGraph coordinates 5 sequential specialized agents communicating over a shared RestaurantAgentState schema:
• **Agent 1 (Classifier):** Maps reviews to 8-category domain taxonomy and computes Severity Score (1.0–5.0).
• **Agent 2 (Root Cause):** Performs hybrid retrieval and maps symptoms to mechanical root cause taxonomy.
• **Agent 3 (Operational SOPs):** Generates 3 prioritized actionable SOPs and an empathetic Guest Recovery Script.
• **Agent 4 (Business Impact):** Computes Churn Risk Tier (Low/Elevated/Critical), annual LTV loss, and star drag.
• **Agent 5 (LLM Quality Judge):** Evaluates feasibility, grounding, tone, and safety compliance, issuing APPROVED or REVISE verdict.

6. Evaluation Strategy (DeepEval, RAGAS & Benchmark Suite)

Evaluated against 8 standardized ground-truth operational benchmark cases (TC_EVAL_01 to TC_EVAL_08) across all categories: **Faithfulness (G-Eval): 94.2%** | **Answer Relevancy: 91.5%** | **Context Precision: 88.7%** | **Context Recall: 91.8%** | **Hallucination Rate: 3.1%**. Real-time telemetry logs end-to-end latency in SQLite with average RRF retrieval under 18.4ms.

7. Challenges, Trade-offs & Engineering Decisions

• **Local CPU Embeddings vs. Cloud APIs:** Traded 1536-dim nuance for zero API costs, sub-10ms local speed, and complete guest privacy.
• **Embedded ChromaDB vs. Distributed Clusters:** Traded multi-node scaling (>500k vectors) for zero-dependency portability on 48,288 records.
• **Bayesian Prophet vs. Deep Learning (LSTM):** Traded multimodal inputs for <1s CPU training, interpretable weekly seasonality, and robust 95% CI bounds on sparse data.
• **Deterministic Heuristic Fallback vs. Pure Generative Output:** Traded natural conversational variance during offline states for 100% operational uptime guarantee.